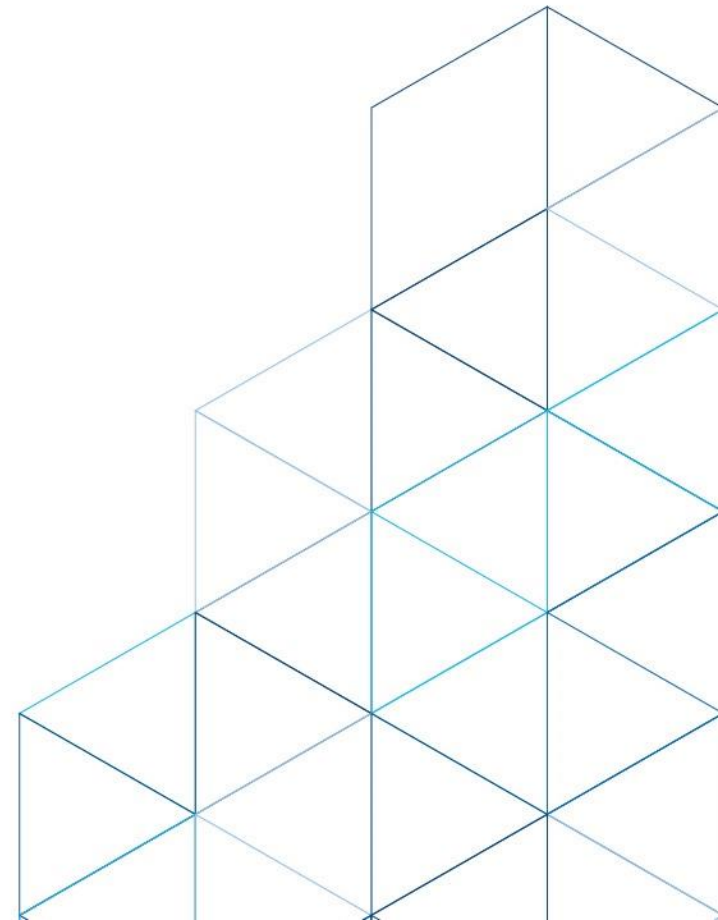


AI Poker Project

DT8042 HT24, Artificial Intelligence
Nov. 2024

Overview

- Introduction to the project
- Timeline
- Five card draw poker game
- Practical information
- Grading and Reporting



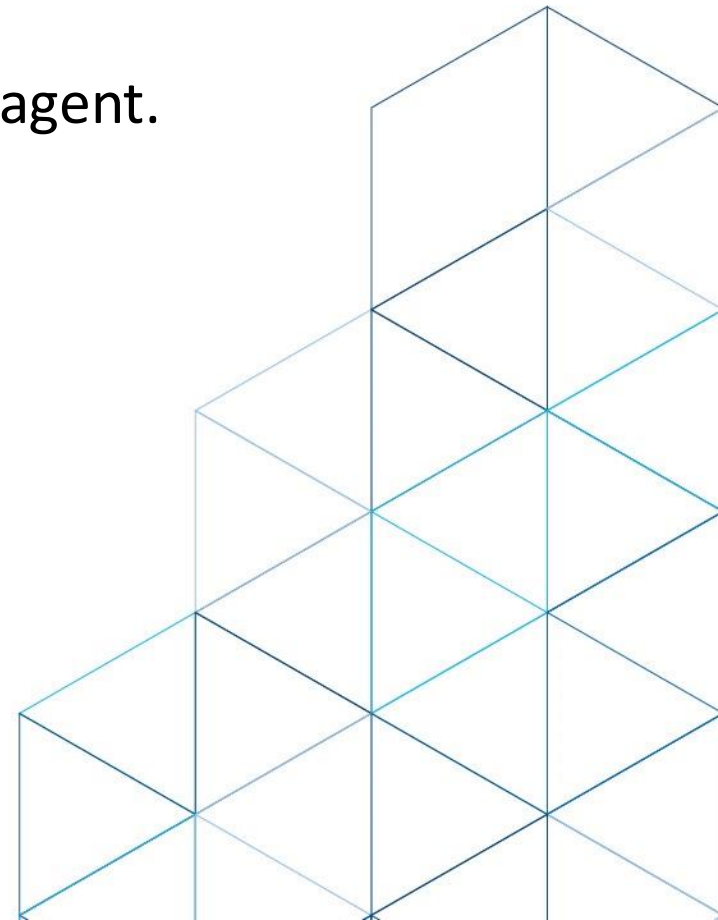
Introduction

- Objective: design an AI poker player for the “five card draw” Poker game
 - It is important to get familiar with the game, e.g. game flow, rules etc.
 - Design strategies for your agent
 - Implementation in Python
- Tournament
 - Registration – from 19th to 29th Nov.
 - 3 members each team recommended
 - Planning for 25 groups in total
 - Two stages
 - Group competition, self-organized, deadline for reporting the results on 6th Jan.
 - Final Tournament, 7th Jan.



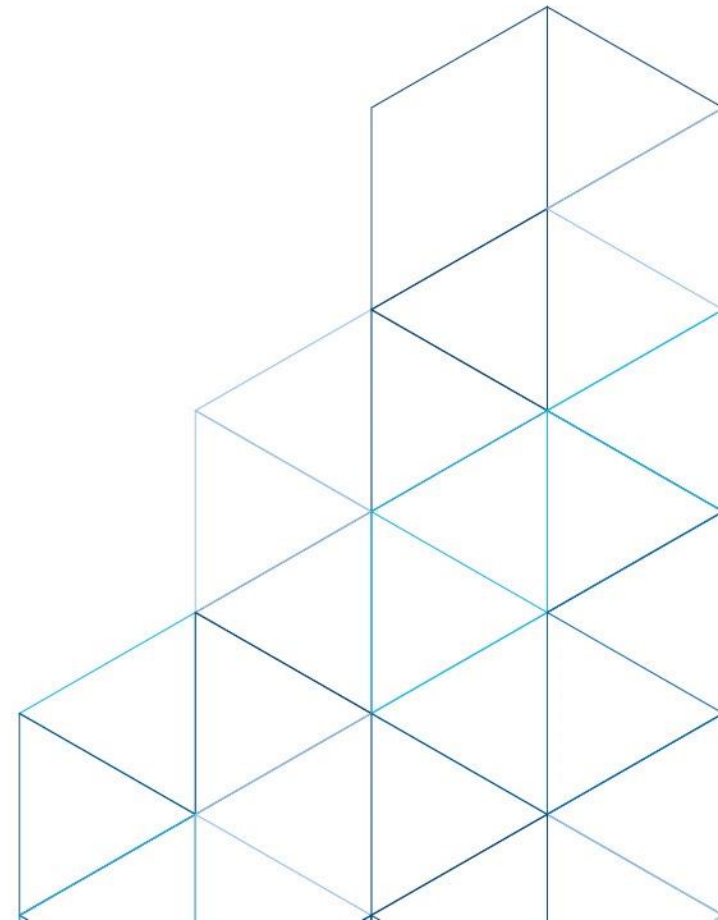
Project Timeline (preliminary)

- Nov. 19th - Team registration open
- Nov. 29th - Project introduction & Registration closed
 - Game flow & rules, poker environment, scripts, log records
- Dec. 4th - Q&A; implementation of agents, e.g. reflex, memory agent.
- Dec. 6th - Q&A; tournament details
- Dec. 12th - Q&A; reporting details
- Dec. 16th – Online Q&A;
- Dec. 19th – Online Q&A;
- Jan. 6th 18:00 – Pre-tournament reporting deadline
- Jan. 7th - Final Tournament
- Jan. 12th 20:00 – Project report deadline



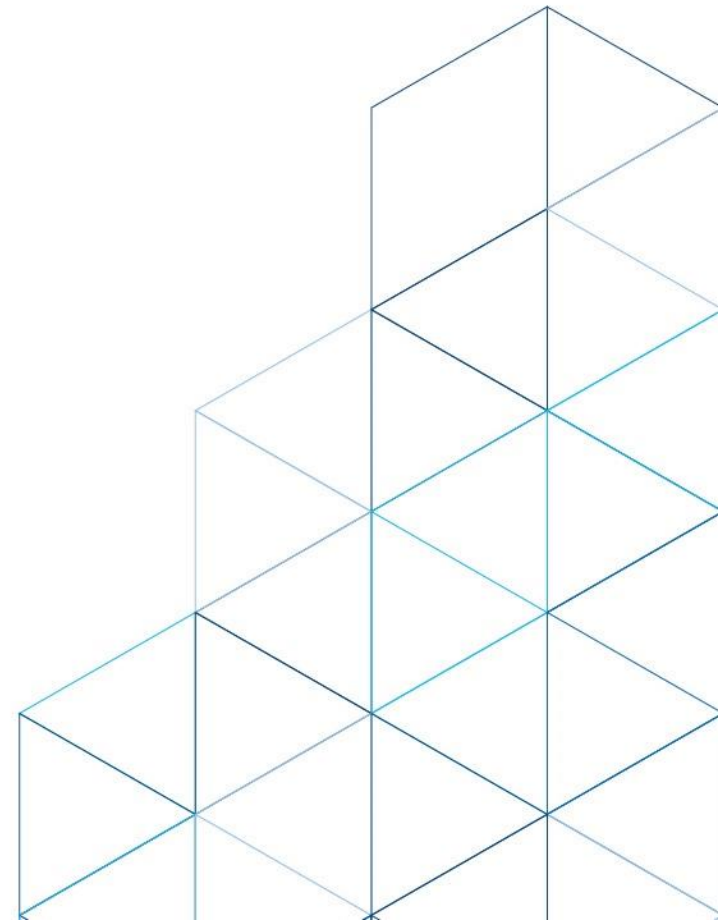
Five Card Draw Poker Game

- 52-card deck, two or more players
- Players bet on the value of the hand (i.e. card combination) and winner is the non-folded player who holds the strongest hand
- Each hand/round contains 7 phases
 - Starting phase - server provide informations
 - Dealling (hands) phase – assign 5 cards to each player
 - Forced betting (ante) phase – all player required to bet for playing this round
 - 1st betting round – open, check, raise, call, all-in, fold
 - Card drawing phase – darw and change card(s)
 - 2nd betting round - check, raise, call, all-in, fold
 - Showdown phase



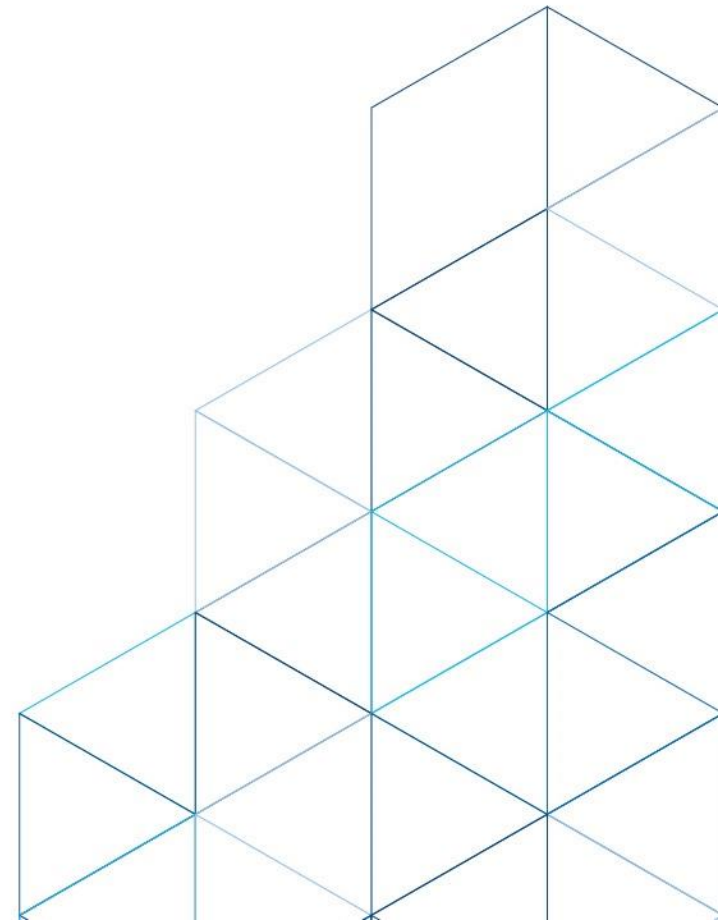
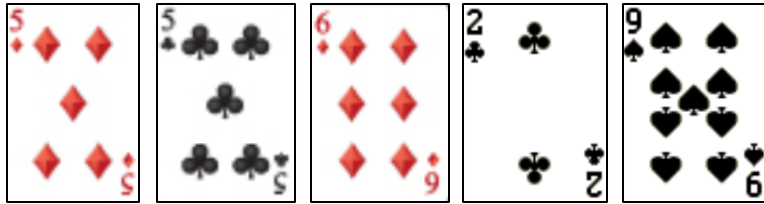
Five Card Draw Poker Game

- Ante
 - A forced bet, a player can only proceed to the first betting phase if an ante is bet
- Betting phase
 - Players actions take in turns, in clockwise rotation
 - The first bet of the game is called “open”
 - Until the first bet is made each player in turn may “check”
 - After the first bet, each player may “fold”
 - “call” to match the highest bet so far made
 - “raise” to increase the previous bet
 - “all-in” bet all coins a player has



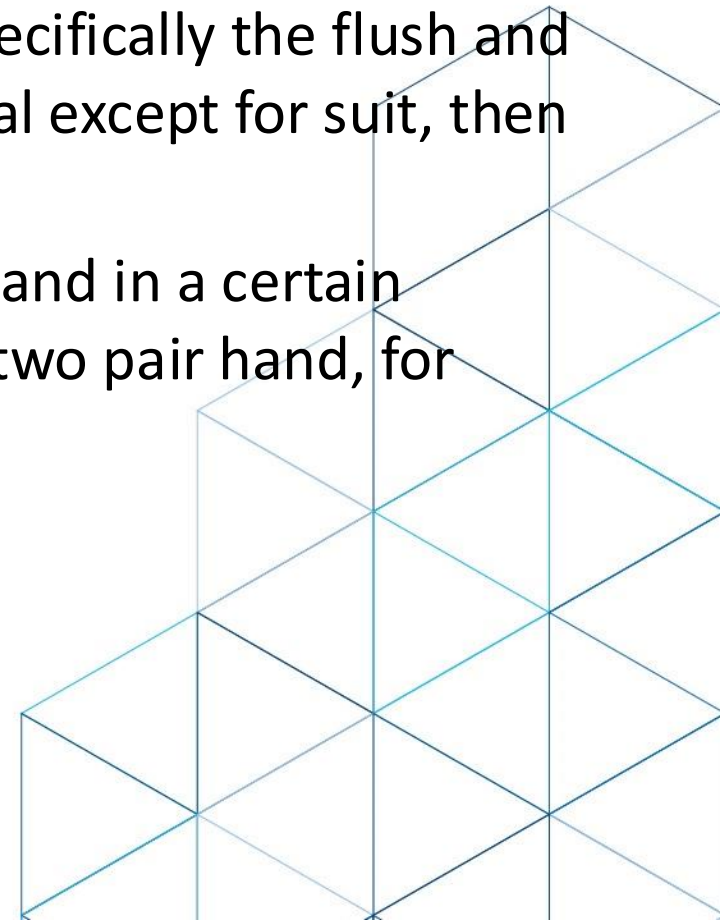
Five Card Draw Poker Game

- Card drawing phase
 - If any players remain in the game, then each player can choose to change cards
 - Select one or more cards for change



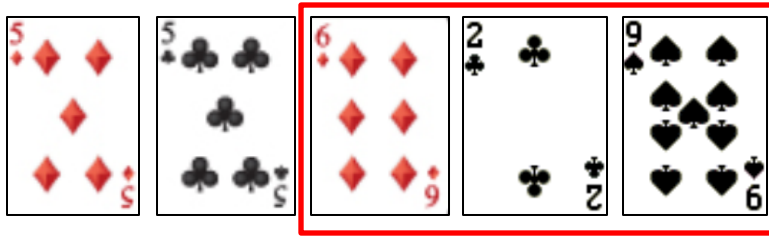
Five Card Draw Poker Game

- Individual cards are ranked A (high), K, Q, J, 10, 9, 8, 7, 6, 5, 4, 3, 2, A. Aces only appear low when part of an A-2-3-4-5 straight or straight flush
- Suits have no value, only for determining certain categories (specifically the flush and straight flush hands). If two players have hands that are identical except for suit, then they are tied and split the pot.
- Hands are ranked by category, and even the lowest qualifying hand in a certain category defeats all hands in all lower categories. The smallest two pair hand, for example, defeats all hands with just one pair or high card.
- Standard poker hands P14



Five Card Draw Poker Game

- Card drawing phase
 - If any players remain in the game, then each player can choose to change cards
 - Select one or more cards for change

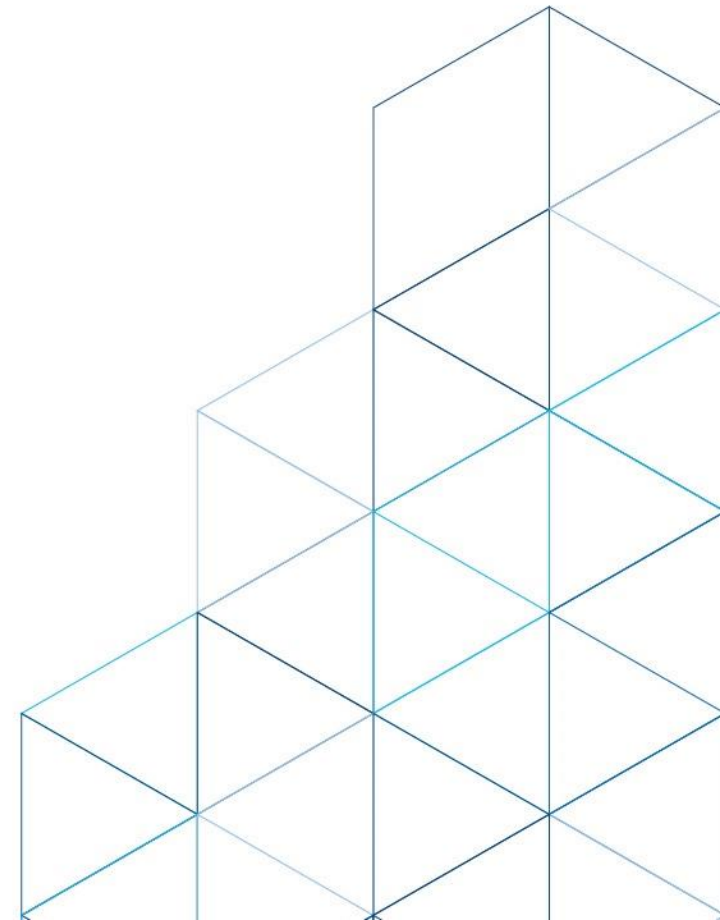


Change



What is the probability of getting...

- Two pairs
- Three of a kind
- Straight
- Flush
- Full house
- Four of a kind
- Straight flush



Practical information

- Poker server and client code

- PokerProject.zip

- PokerServerGUI – server program hosts the game
 - PokerClientRandom – client code
 - PokerGame.py - communication with the server
 - Client.py – Implement your agent in this file

- Poker project Rules

- AI_poker_project.pdf - P2, 12-15

- The rest (P3 – P11) is only useful for programming in java and execute on a phone emulator



Grading & Reporting

- Grade
 - Grade 3: a reflex agent that takes its own hand and/or the opponent's action into account.
 - The strategy should be well-motivated
 - A comprehensive analysis based on experiments and group competition
 - Grade 4: an agent with learning capability in-built, utilizing historical data
 - Grade 5: an agent with sophisticated learning capability and shall achieve good results in the tournament.
 - The approach shall be very well motivated and documented.
- Submit your project files on the Blackboard
 - Report
 - Introduction, method, result and conclusion
 - Code

